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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

REAL TIME RESEARCH PROJECT

Batch: 15

Team details:

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Title: MIO - A Multipurpose AI assistant

Abstract:

Mio is an advanced, React-based AI assistant designed to deliver an intuitive and intelligent user experience by seamlessly integrating Google Gemini for powerful natural language processing and code generation capabilities. Mio offers a multifaceted suite of features including Chat AI for conversational interactions, Code AI for generating and explaining code snippets, speech-to-text functionality for hands-free communication, markdown support for rich text formatting, and real-time responsiveness that ensures fluid and engaging interactions. The platform leverages a sleek and modern Material-UI interface, enhanced by dynamic loading animations, intuitive error handling, and built-in speech recognition to foster ease of use and accessibility for users across varying levels of technical expertise.

Built on a robust tech stack of React for the front-end and Express.js for the back-end, Mio is engineered for scalability, high performance, and adaptability, making it suitable for both everyday users and developers seeking efficient and responsive tools. Its modular architecture facilitates easy integration and extension of features, while maintaining optimal performance and low latency. The name "Mio," meaning "my" in Italian, underscores its personalized, user-centric design philosophy — positioning the assistant not merely as a tool, but as a reliable and intelligent companion that adapts to the needs of its users. With its comprehensive set of capabilities and seamless interactions, Mio stands as a forward-thinking AI assistant that bridges communication, productivity, and innovation in one unified platform.

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